SmartHire - Final Project Report

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1. Project Overview

SmartHire is a resume-to-job matching portal built using classical

2. Datasets Used

Dataset	Size	Purpose
Resume Dataset	962 resumes, 25 categories	Classifier training
Naukri Job Listings	30,000 jobs	Job corpus
LinkedIn Job Postings	118,000+ jobs	Job corpus
Total Jobs	148,718 jobs	Recommender

3. System Architecture

Resume Upload (PDF/DOCX)

Streamlit Web Portal

4. Machine Learning Models

Model A - Resume Category Classifier (Supervised)

- Algorithm: Logistic Regression

Metric	Score
Accuracy	~99%
Precision	~1.00
Recall	~1.00
F1-Score	~1.00

- Observation: The classifier performs excellently across all

Model B - Job Recommender (Unsupervised)

- Algorithm: TF-IDF + Cosine Similarity

Model C - Job Clustering (Unsupervised)

- Algorithm: KMeans (k=8)

Model D - Skill Gap Report (Unsupervised)

- Method: Compare resume skills vs top skills in matched jobs

5. Results Summary

Component	Method	Result
Resume Classifier	TF-IDF + LR	99% accuracy
Job Recommender	Cosine Similarity	Relevant top-10 jobs
Job Clustering	KMeans (k=8)	8 distinct job families
Skill Gap	Cluster Analysis	Missing skills identified

6. Web Portal Features

- Upload PDF or DOCX resume

7. Limitations

- Datasets are from 2019-2024, may not reflect current job market

8. Future Improvements

- Replace TF-IDF with sentence embeddings for better matching

9. Tools & Technologies

- Python 3.14 - Core language

10. Repository

GitHub: https://github.com/pujalokku49-stack/smarthire

11. Conclusion

SmartHire successfully demonstrates a complete end-to-end ML pipeline